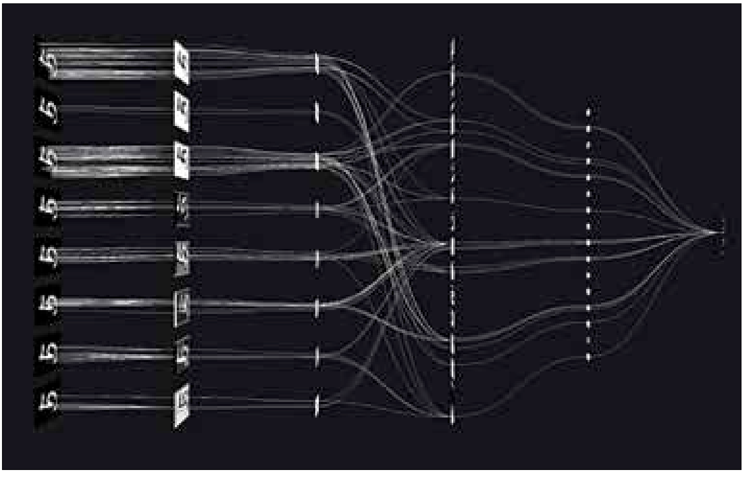
multiple convolutional layers, pooling layers, some fully connected layers and normalisation layers. Convolutional networks are so desired by the industry due to the increased efficiency they present for image processing and natural language processing which we'll see in the next segments.

HOW DO THEY WORK

CNNs use a pooling operation with a max-pooling layer that selects the maximum value from a group of features of the inputted content. Just like a convolutional layer, all the pooling layers are parameterised dependent on the size of the image.

Take the example of an image that is inserted at the input. The system breaks down the image into a series of rows and columns that are rearranged and reassembled as they go through the different nodes and layers. These pooling layers help to scale down the dimensionality by retaining only the details that are important. But here's the catch, even if the input is shifted or changed regarding its pixel size, orientation or space, the system will select the maximum number of values. Pooling layers are usually kept hanging between successive convolutional layers.

So that's pretty much it, all CNNs have convolutions that extract features from local regions of the inputted content. You'll find a combination of pooling, affine and cross-linked layers. CNNs have thus become the staple resource for studying audiovisual content due to their cross-linking archi-



Learning variations of the same object